UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 009 · Introduction to maxima and minima for a function of two variables · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 ("Extremas of functions"), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c

Contents

1. 1 · Maxima, minima and saddle points
2. 2 · Definitions: relative maximum, minimum and extreme value
3. 3 · Conditions and the working rule
4. Past-paper questions

1 Maxima, minima and saddle points

Definitions

Saddle point. A stationary point at which $f(a,b)$ is neither a maximum nor a minimum: the surface rises in some directions and falls in others. Test: $rt - s^2 < 0$. S009·5 · S009·7 · T009 00:11:06

Key points

1. One variable recap: solve $\frac{dy}{dx} = 0$ for critical points; $\frac{d^2y}{dx^2} < 0$ gives a maximum, > 0 a minimum; $= 0$ gives no conclusion. The tangent at a maximum or minimum is parallel to the x-axis (slope 0). ESA ★★ T009 00:00:50 · T009 00:01:40 · T009 00:02:57
2. In two variables the tangent plane at a maximum or minimum is horizontal (parallel to the xy-plane); slide 009·3 shows a horizontal tangent plane at a relative maximum (Figure G1.4). ESA ★ S009·3 · S009·4 · T009 00:01:10
3. A function can have several local maxima and several local minima (slides 009·3–4: "These are all local maxima / minima"). ESA ★ S009·3 · S009·4 · T009 00:00:16

Exam traps

Exam trap. In one variable $\frac{d^2y}{dx^2} = 0$ is not "a saddle point"; it is only inconclusive (x^4 has a minimum, x^3 an inflection). ADDED FOR CLARITY T009 00:03:14

2 Definitions: relative maximum, minimum and extreme value

Definitions

Relative maximum. A function $f(x, y)$ is said to have a relative maximum at a point (a, b) if there exists a neighbourhood of the point (a, b), say $(a + h, b + k)$, h and k small, such that $f(a, b) > f(a + h, b + k)$. S009·6 · T009 00:03:46

Relative minimum. If $f(a, b) < f(a + h, b + k)$ then $f(x, y)$ is said to have a relative minimum at (a, b). S009·6 · T009 00:04:58

Extreme value. $f(a, b)$ is said to be an extreme value of f if it is either a maximum or a minimum. S009·6

Neighbourhood. The points very close to (a, b): $(a + h, b + k)$ with h, k small. "Relative" means largest or smallest only in such a neighbourhood, not over the whole plane. T009 00:04:12 · T009 00:04:40

Statements in the class material

> *"In other words if $\Delta = f(a + h, b + k) - f(a, b)$ is of the same sign for all small values of h, k and if this sign is negative then $f(a, b)$ is maximum & if this sign is positive then $f(a, b)$ is a minimum."* S009·6 · T009 00:05:23

Key points

1. "Relative" because only points in a neighbourhood of (a, b) are compared, not the whole plane. ESA ★ T009 00:04:40

3 Conditions and the working rule

Definitions

Stationary (critical) point. A point where $f_x = 0$ and $f_y = 0$: the tangent plane is horizontal, and the function may have an extreme value there. S009·8 · T009 00:02:47 · T009 00:06:19

Hessian. The determinant $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = rt - s^2$ of second partial derivatives. S009·7 · T009 00:07:22

Statements in the class material

> *"The necessary and sufficient condition for $f(x, y)$ to have a maximum value at (a, b) is $f_x(a, b) = 0$, $f_y(a, b) = 0$ and $rt - s^2 > 0$, where $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. $rt - s^2$ is the determinant of the Hessian matrix $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}$."* S009·7

> *"Note: If $rt - s^2 < 0$, $f(a, b)$ is not an extreme value, i.e. $f(x, y)$ is neither a maximum nor a minimum at the point (a, b) and the point is called a saddle point. If $rt - s^2 = 0$, then the test provides no information about the nature of the point."* S009·7

Key points

1. Working rule step 1: find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, equate each to zero, solve simultaneously; let $(a, b), (c, d), \ldots$ be the pairs of values. ESA ★★★ S009·8 · T009 00:11:32
2. Step 2: calculate $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$ for each pair of values. ESA ★★★ S009·8
3. Steps 3–6: $rt - s^2 > 0$ and $r < 0$: maximum; $rt - s^2 > 0$ and $r > 0$: minimum; $rt - s^2 < 0$: not an extreme value, (a, b) is a saddle point; $rt - s^2 = 0$: the case is doubtful and needs further investigation. ESA ★★★ S009·8 · S009·9 · T009 00:12:33
4. Two important conditions to remember (lecture): both first partials zero, and $rt - s^2$ from the second partials. ESA ★★★ (transcript only) T009 00:08:50
5. Unlike one variable, the sign of r alone is not enough: first check $rt - s^2 > 0$, then look at r. ESA ★★★ T009 00:07:51 · T009 00:09:43
6. The value of the function at a maximum or minimum point is the maximum or minimum value: always substitute the point back into f. ESA ★★ (added for clarity) T009 00:04:51
7. Each stationary point is tested separately: a function can have maxima, minima and saddle points together (class 010: four points for $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$). ESA ★★★ ESA Dec-23 1c · ESA Jan-24 s1 1a

Why?

Why must f_x and f_y both vanish at an extremum?
Along the line $y = b$, $f(x, b)$ is a function of x alone with an extremum at $x = a$, so its derivative $f_x(a, b) = 0$. The same along $x = a$ gives $f_y(a, b) = 0$. Geometrically, the tangent plane is horizontal.

(added for clarity) T009 00:06:19

Why does $rt - s^2 < 0$ mean a saddle?
Near a stationary point $f(a+h, b+k) - f(a,b) \approx \frac{1}{2}(rh^2 + 2shk + tk^2)$ (Taylor, class 007). This quadratic keeps one sign for all h, k exactly when $rt - s^2 > 0$ (sign of r); when $rt - s^2 < 0$ it is positive in some directions and negative in others.

ADDED FOR CLARITY S009·6 · S009·7

Why is r enough to separate maximum from minimum when $rt - s^2 > 0$?
$rt > s^2 \geq 0$, so r and t have the same sign: the surface curves the same way along x and along y.

ADDED FOR CLARITY S009·8

Exam traps

Exam trap. $rt - s^2 < 0$ is a saddle point (a definite answer); only $rt - s^2 = 0$ is "no conclusion". The lecture mixes the two at one point. S009·7 · T009 00:08:23 · T009 00:11:06

Exam trap. Do not decide maximum or minimum from r before checking $rt - s^2 > 0$: $x^2 - 3xy + y^2$ has $r = 2 > 0$ at the origin but $rt - s^2 = -5$: a saddle. ADDED FOR CLARITY S009·8

Exam trap. $f_x = f_y = 0$ is necessary, $rt - s^2 > 0$ sufficient; slide 009·7 calls the pair "necessary and sufficient". ADDED FOR CLARITY S009·7

Exam trap. Evaluate r, s, t at each stationary point; a sign that holds at one point need not hold at another. S009·8

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. The definitions are never asked by themselves, but the test of this session is the whole of three 7-mark Q1 parts: Dec 2023 Q1.c and Jan 2024 set 1 Q1.a (the same function $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$) and Jan 2024 set 2 Q1.a ($x^3 - 12x + y^3 + 3y^2 - 9y$). They are worked in the class 010 pack; the method is starred ★★★ here.

No past-paper questions in scope.